

An Emerging Approach to Look Development Using Reverse Engineering and Mind Mapping in Animation Education Centered on Entertainment-Related 3D Computer Graphics in Japan

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Abstract

In the Japanese entertainment industry, the field of 3D computer graphics (CG) animation is characterized by a diverse range of stylistic approaches, encompassing both traditional 2D aesthetics and Western photorealistic 3D representations. Conventional 3D CG education has focused on the acquisition of technical skills, with less emphasis on the development of creativity and project-realization skills, which are essential for addressing the diverse range of visual styles observed in the industry. This study proposes a new educational approach that combines reverse engineering, mind mapping, and project-management techniques to compensate for deficiencies in traditional 3D CG education. With this approach, students engage in an analytical observation of existing works across a broad spectrum of visual styles, identifying both their technical and artistic elements. This is achieved through the use of mind mapping, which facilitates the generation of original ideas. This process enables the integration of established styles with individual creativity, thereby encouraging divergent thinking. Subsequently, the concept art is translated into a practical production using digital content creation (DCC) tools. At this juncture, project-management techniques are introduced to validate the feasibility of the ideas and learn efficient production processes. Through critical thinking, students enhance their creativity, technical proficiency, and practical management skills. This study aims to establish a comprehensive educational methodology that combines creativity that is responsive to diverse styles, advanced technical skills, and practical project-management abilities. This approach is expected to foster innovative and feasible visual art projects, contributing to the overall advancement of the entertainment industry.

CCS Concepts

• Social and professional topics; • Professional topics; • Computing education;

Keywords

3D Computer Graphics (3D CG), Reverse Engineering, Mind Mapping, Project Management, Creativity Enhancement, Technical Skills, Visual Style Diversity, Critical Thinking, Entertainment Industry Advancement, Educational Methodology

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1 Introduction

Japanese animation is globally acclaimed for its unique visual style, which is rooted in the fusion of traditional Japanese aesthetics and contemporary culture. As Suan, 2013 pointed out, Japanese animation is characterized by aesthetic expressions influenced by ukiyo-e and Japanese painting. Gan, 2009 noted that the expressive character designs typical of Japanese animation, which feature large eyes and delicate lines, create strong emotional connections with audiences worldwide. The evolution of 3D CG technology has diversified Japanese animation expressions. The toon shading technique described by Vanderhaeghe et al., 2011 enables hybrid expressions that combine hand-drawn animation and 3D CG; this technique has been widely adopted not only in animation but also in the gaming industry. As Yoon and Kim, 2012 observed, Japanese studios actively incorporate new technologies, integrating traditional methods with digital techniques to produce innovative expressions across various fields, including Visual Effects (VFX), VTubers, music, and video streaming. However, educational methods for analytically understanding the technical and artistic elements of these diverse expression styles and utilizing this knowledge to create original works are not sufficiently established. In particular, there is a need for an approach that combines the cultivation of creativity with the development of practical skills to turn ideas into concrete works.

This study proposes a novel educational methodology that integrates reverse engineering, mind mapping, and project-management techniques. In this approach, students first gain an analytical understanding of diverse styles through reverse engineering. They then generate original ideas using mind mapping while simultaneously utilizing project-management techniques to

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complete these ideas as feasible works. This method aims to comprehensively develop analytical, creative, and practical skills. To achieve this objective, the following main research question has been identified:

How does a novel educational approach integrating reverse engineering, mind mapping, and project-management techniques enhance students' creativity, technical proficiency, and practical skills in Japanese entertainment-related 3D CG animation education?

The following subsidiary research questions will also be asked:

1. How does the analysis of existing works using reverse engineering influence students' technical understanding and artistic insights?
2. How does the idea-generation process based on mind mapping contribute to the improvement of students' creativity and innovation capabilities?
3. How do students' teamwork abilities and practical problem-solving skills develop during a CG production process that integrates forward engineering and project-management techniques?

2 Background and Related Work

2.1 Look Development in Animation

Look development in animation production refers to the process of developing textures and lighting; it is essential for maintaining visual consistency and realizing the director's vision. This process enables the unification of the artistic direction of the production, allowing for a consistent visual experience for the audience Thompson et al., 2021 emphasized the importance of experimental-look development by examining the production process of Pixar's short animation *OUT*. Jordan et al., 2022 presented a highly artistic approach focused on materials and directorial methods, including character design and environmental textures, based on their study of the production of *Lightyear*.

2.2 Educational Approaches to Look Development

The importance of look development is well recognized in art universities and specialized schools in the United States. The Ringling Academy (2024) and Gnomon, 2024 offered look development curricula in their advanced classes, while the Savannah College of Art and Design, 2024 provided similar curricula within their game development program.

In contrast, while some departments in Japanese universities and specialized schools are engaging with 3D CG animation and game development, educational methods for analytically understanding and reproducing ideation techniques and project management in the context of CG entertainment remain limited. Given this situation, it is crucial for Japanese educational institutions to recognize the importance of look development in project management and introduce curricula that deal with it. This would enable students to acquire the advanced management skills and creativity demanded by the industry, potentially developing them into globally competitive professionals.

2.3 Creative Design Processes in Entertainment

In the field of design, Gonçalves et al., 2014 investigated designers' preferred inspirational approaches during idea generation and analyzed the influence of various sources of inspiration. The creative design process that inspires designers is complex and multifaceted. When analyzing the creative process in animation projects, there is a role for creativity plays a crucial role at multiple stages, including initial concept development, storyboarding, modeling, and animation production, which requires feasible, multifaceted idea creation. Furthermore, Tschang, 2007 examined the design process in the video game industry, and they highlighted the importance of exercising creativity within technological constraints.

2.4 Reverse Engineering in Design Education

The reverse engineering method is a technique aimed at analyzing existing products or systems to understand their design and functionality in the engineering field (Chikofsky and Cross, 1990). In the context of education, Dym et al., 2005 and Dalrymple et al., 2011 reported that reverse engineering contributes to the improvement of students' analytical skills, problem-solving abilities, creativity, and critical thinking.

Furthermore, Matsunaga and Miyata, 2024 have proposed an approach to deepen the understanding of visual storytelling techniques by analyzing subjective, objective, and artistic elements from existing concept art. In the present study, we identify and combine various elements that characterize visual expressions from multiple entertainment works to create new styles. Using forward engineering, we also produce a masterpiece to create actual 3D CG works.

2.5 Exploring Personal Organs Through Mind Mapping

In this study, we introduce the creation of mind maps dealing with a theme called My Primal Landscape (the landscape in one's mind's eye) for original project planning. This method is based on the model presented in Feinstein's (2023) "Creativity in Large-Scale Contexts," which focuses on the relationship between personal experiences and the manifestation of creativity. The exploration of the My Primal Landscape theme is expected to produce two main effects:

1. A refinement of ideas. In this case, students are expected to develop ideas derived from their primal landscapes through dialogue and collaboration with others. This aligns with the process of idea development through social interaction emphasized by Feinstein, 2023.
2. A nurturing of originality. By recognizing their patterns of creative thinking and past significant sensory experiences, and by consciously incorporating these into their work, students can cultivate the ability to produce unique expressions.

3 Proposed Method

The curriculum developed as part of this research (for the syllabus, see Supplement Table 1) centers on a look development method that fuses reverse engineering and mind mapping. The curriculum consists of the following four steps:

Step 1: Proposal Submission and Approval. Students create visually oriented proposals and obtain approval from instructors.

Step 2: Visual Reference Collection and Initial Asset Creation. Based on the approved proposal, teams collect references and create initial assets.

Step 3: My Primal Landscape and Concept Art Creation. Students create unique concept art by combining personal experiences with their references.

Step 4: Feasibility Assessment Using DCC Tools. The feasibility of CG production is evaluated by the instructors based on the created concept art.

While these steps proceed in order, students can go back to previous steps as needed for iterative improvement. An overview of Steps 2–4 is shown in Supplement Figure 1, and a specific example flow of Steps 2–3 is found in Supplement Figure 2.

Students will adopt the approach proposed by Matsunaga and Miyata, 2024, aiming to merge the analysis of existing works' world-views, stories, colors, contour lines, and character styles with their personal primal landscapes.

3.1 Step 1: Submission and Approval Of Graduation-Project Proposal

Students submit visually oriented project proposals, which include style images for characters, backgrounds, colors, and visual effects, as well as a story overview and a team member structure (see Supplement Figure 3). The team structure is hierarchical, with core members at the top; the leader serves as both producer and project manager, followed by the director, art director, CG supervisor, and line producer (LP).

The CG supervisor oversees comprehensive CG methodologies; together with the other two core members, they supervise the artist supervisors. Each supervisor guides and manages their group of specialized artists.

The core members are appointed by the producer, and there can be overlapping roles within projects, but not between projects. For instance, the director may also serve as the producer, or the CG supervisor might also be the supervisor of a specific department. Artists can participate in multiple projects, maximizing the use of human resources. The LP collaborates with artists via the CG supervisor and other supervisors, managing the project's progress and providing support both technically and procedurally. Instructors evaluate the role sharing and skills of each student, judging the feasibility of the proposal before approval. After approval, in addition to the supervising instructor, students can select a primary instructor from among the five instructors involved in the graduation project. For team-size allocation, refer to Supplement Table 2.

3.2 Step 2: Collecting References for Approved Concepts and Creating Initial Assets

After approval, the entire team is divided into multiple teams based on roles. The main teams include:

1. The story development team (scriptwriters and storyboard artists)
2. The character-asset development team (character modelers, riggers, etc.)

3. The test sequence team (background, props, effects, layout, and lighting specialists)
4. The production management team (schedule management and resource allocation)

This classification, which is shown in Supplement Table 3, is a reorganization of the eight artist groups presented in the organizational chart of Supplement Figure 3, grouped by specialized skills. Each team collects reference images and conducts reverse engineering. Students with multiple specialized skills can take on multiple roles or join different teams, thus solving talent shortages and increasing the flexibility and efficiency of the whole team. The entire team collaborates on creating a comprehensive image board, concept art, and scale models. Progress will be added to the image board and shared as needed.

3.3 Step 3: The My Primal Landscape Theme and Concept Art Creation

In this stage, the director creates mind maps based on the My Primal Landscape theme sheet (Supplement Table 4) to explore the theme in depth. Through critical thinking sessions with the supervising instructor and the primary instructor, students aim to merge their references with mind maps to stimulate creative ideas (this procedure is detailed in Supplement Table 5). The resulting creative and technical elements are combined and refined to clarify specific aspects, such as contour lines, shaders, lighting, color schemes, visual effects, and the degree of matching between character designs and backgrounds. These refined visual and technical specifications are used to develop pivotal scene concepts by tracing and overpainting 3D data (Supplement Figure 2, right). These scene concepts serve as both visual benchmarks and technical proofs of concept. They are designed to be key visuals that will guide the CG implementation, validate the technical feasibility of the project, and demonstrate the core aesthetic and narrative elements of the proposed animation.

3.4 Step 4: Feasibility Assessment Using DCC Tools

This curriculum, codeveloped with two CG production companies, reflects the latest industry needs for practical project operations (for the production workflow up to compositing and the approval system, see Supplement Figure 4). In the final stage, thanks to forward engineering methods, the technical elements are deconstructed and reconstructed in the process of transforming the concept art into CG works. During this process, feasibility is assessed by evaluating both the rendering time of the prototype scene (referred to as the "master shot" in Supplement Figure 1, right) and its development duration. This is crucial as visually impressive concepts may be impractical due to excessive rendering times. This assessment also considers the complexity of the animations and visual effects, as well as the team's technical capabilities.

4 Evaluation Methods

This study adopted a mixed methods approach to evaluate the effectiveness of the proposed methodology. The subjects were 10 teams consisting of 120 third-year students (70% male, aged around 20) from a three-year CG designer course at a Japanese vocational

school. Data collection was approved by the educational institution and conducted based on ethical principles.

To collect quantitative data, an adapted version of the California Critical Thinking Skills Test (CCTST) was used. This test is widely recognized as a reliable tool for evaluating critical thinking (Facione et al., 2002). Pretests and posttests were conducted on the 10 teams to assess the effectiveness of the proposed educational method. Seven skills were assessed (analysis, reasoning, explanation, evaluation, deduction, teamwork, and creativity and innovation) on a scale of 1 to 5, with a maximum total score of 35 (the evaluation table is shown in Supplement Table 6). Qualitative data collection involved semi-structured interviews with project planners. This method is suitable for exploring participants' experiences and understanding in depth (Kallio et al., 2016). Furthermore, all the participants were asked to record their production processes using social networking services (Slack, Discord, etc.) to track the development of technical understanding, creative thinking, and problem-solving during the learning process. Insights into student growth, challenges, and interactions were also collected through in-class observations recorded by the instructors.

Correlation analysis and multivariate analysis were used for the quantitative data. Thematic analysis (Braun and Clarke, 2006) and content analysis (Erlingsson and Brysiewicz, 2017) were employed for the interview data. Furthermore, based on the convergent parallel design of mixed methods research (Creswell and Plano, 2018), the results of the quantitative and qualitative data were combined to comprehensively examine the effects, mechanisms, and areas for improvement of the proposed method. Finally, the approach of this study was independently evaluated by experts from external companies during face-to-face interviews.

5 Results

Correlation and multivariate analyses were applied to the pretest and posttest scores obtained with the modified version of the CCTST. Correlation analysis revealed moderate positive correlations ($r = 0.45\text{--}0.60$, $p < 0.01$) between improvement in reverse engineering skills (analytical ability) and improvement in other skills (reasoning, explanation, evaluation, and deduction).

Regarding the multivariate analysis, multiple-regression analysis was conducted with mind-mapping utilization as the independent variable and the creativity and innovation scores as the dependent variables. The results show that mind-mapping utilization was a significant predictor of creativity ($\beta = 0.38$, $p < 0.05$). A notable effect of the proposed educational method was that dialogue with instructors, including guidance in two critical thinking sessions, significantly contributed to the students' insights. The results of the pretests and posttests are shown in Figure 1 (detailed scores are provided in Supplement Table 7). As an effect of this initiative, improvements in overall scores were observed in 9 out of 10 teams. Team I and Team J showed particularly remarkable improvements, recording increases of 15 and 13 points, respectively. These teams were able to reflect deeply on their analytical and creative processes and identify areas for betterment through dialogue and critical thinking guidance.

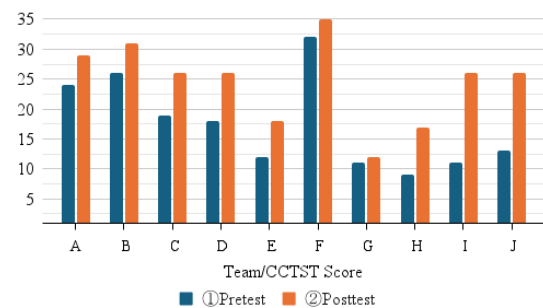


Figure 1: Results of the pretests and posttests using the CCTST

The impact of analyzing existing works using reverse engineering methods on students' technical understanding and artistic insights was evident in the improvement or maintenance of the analytical ability scores of all the teams. For instance, Team I showed a significant improvement in analytical skills (from 1 to 4 points). This team gained insights by using critical thinking, understanding the essential structures and techniques of works rather than merely imitating them, and applying this to their productions. Most teams showed similar improvements. However, cases such as Team G, which had low initial scores and limited improvement (from 11 to 12 points), were also observed. Regarding the effectiveness of the idea-generation process with mind mapping, 7 out of 10 teams showed improvement in creativity and innovation ability.

The interviews with students who effectively utilized mind maps revealed interesting insights. These participants closely related personal experiences and emotions, including childhood experiences, memories from student life, and dreams, to their works. When explaining the driving forces behind their creations, some of these participants said that childhood-home experiences were at the root of their compositions; others spoke of wanting to overcome memories of failures from their student days or recreate things seen in dreams. This phenomenon aligns with McAdams, 1993 theory of the reconstruction of self-narrative. The students attempted to reconstruct their self-narratives by reinterpreting past experiences and deep subjective sensations through their works, thus expressing them in new contexts. This process evoked strong desires for creation, and it played a crucial role in motivating creativity.

The instructors' observation logs contain statements such as "Knowing and sharing the students' deep motivations helps me better understand their intentions" and "I am impressed by the students' natural analogical integration and surprisingly rich creativity." These statements suggest that critical thinking through mind mapping can deepen the dialogue between students and instructors, potentially leading to more appropriate guidance. However, phrases such as "I don't want to reveal my personal opinions" and "I don't want to expose my past to others" were also heard. As pointed out by Gonçalves et al., 2014, these results indicate that analogical idea-generation methods can be challenging for some students.

Still, in Japanese animation production, these approaches can be effective means of achieving originality. Okada (2006) pointed out the effectiveness of analogically combining team members' creativity, as seen in Anno et al. (1987) regarding the film *Royal Space Force: The Wings of Honnêamise*. Moreover, Miyazaki's 2023 movie *The Boy and the Heron* is a compelling example of the unique creativity of Japanese animation as well as of how reconnecting with one's unconscious memories contributes to self-discovery. This film embodies Miyazaki's philosophy that engaging with latent memories and emotions can generate novel perspectives and foster creativity. Doing so can also enhance the distinctive nature of Japanese animation. This approach adds depth to narratives and aligns with the educational method proposed in this study, which suggests the importance of originality in nurturing the next generation of creators.

However, the findings indicate that the effectiveness of the idea-generation process that uses mind mapping largely depends on the student's attitude toward creation. This suggests that the impact of the proposed method will vary based on students' dispositions and approaches to the creative process. In particular, a more flexible and individualized approach may be necessary for pupils with low skill levels or those who are resistant to incorporating their personal experiences into their work. Future research may expand the effectiveness and scope of this approach, potentially leading to the realization of more comprehensive creativity education.

The development of students' technical skills, creativity, and practical abilities in the CG production process was confirmed by improved teamwork scores in eight teams and improved evaluation skills (reflecting practical problem-solving ability) in seven teams. Notably, Teams D and I showed significant improvements in analysis, explanation, evaluation, and deduction skills (from 2 to 4 points). During the critical thinking sessions, deep discussions were held on team role sharing, progress management, and problem-solving approaches, which likely resulted in the improvement of these skills. These conclusions highlight the potential of the proposed method for fostering creativity and practical skills in animation education; they also underscore the need to apply the method in a flexible and personalized manner. The outcomes of the method provide valuable insights into the future development of creative education in the context of Japanese animation and CG production.

6 Corporate Evaluation and Industry Perspective

To evaluate the effectiveness and practical applicability of our educational approach, we conducted face-to-face interviews with experts from four companies. Overall, the evaluation of the proposed method was positive; the method was seen to align with the industry's needs (see Table 8 in the Appendix for the interview summaries). Two comments stand out as particularly significant.

Company A highly praised the creativity-fostering effect of the proposed method, stating, "It is very effective in allowing students to imbue originality into what they want to create." This testimony supports the efficacy of our approach to bringing out students' originality.

Company D emphasized the importance of this method in the era of AI, saying, "In the AI era, it is very important to conduct

basic education that fosters fundamental creative skills, such as determining what one wants to do." This observation suggests that the proposed method not only meets current industry needs but also contributes to the development of fundamental skills that can adapt to future technological changes.

These evaluations indicate that our educational method aligns with the industry's requirements and may lay the foundations for future creativity education. At the same time, it was suggested that further verification is needed regarding the method's practical application and long-term effects.

7 Conclusions

This study proposed a look development method that integrates reverse engineering and mind mapping, and evaluated its effectiveness using an explanatory mixed methods approach. The analysis showed that this method effectively enhanced creativity, critical thinking, technical and artistic skills, and problem-solving abilities in CG production. In addition, it fostered teamwork and improved students' technical understanding and artistic insights, with the fusion of reverse engineering and mind mapping facilitating the creation of unique visual styles.

To address the main research question, the proposed method demonstrated that combining the analysis of existing work with individual creativity significantly enhances students' technical proficiency and originality. The reverse engineering process deepened students' technical understanding and artistic insight, while mind mapping boosted their creativity and innovation skills. Moreover, integrating project-management techniques into the workflow improved students' teamwork and practical problem-solving abilities.

However, this study was limited by its small sample size and single-institution focus, which hindered broader comparisons. Future research should explore the application of the method in other settings and consider providing additional support for students who need more time to master these techniques. Further studies are also needed to examine the long-term effects of the approach.

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